

Dataset Description and Statistical Overview

Montgomery County, Shenzhen, and Darwin Chest X-ray Lung Segmentation Datasets

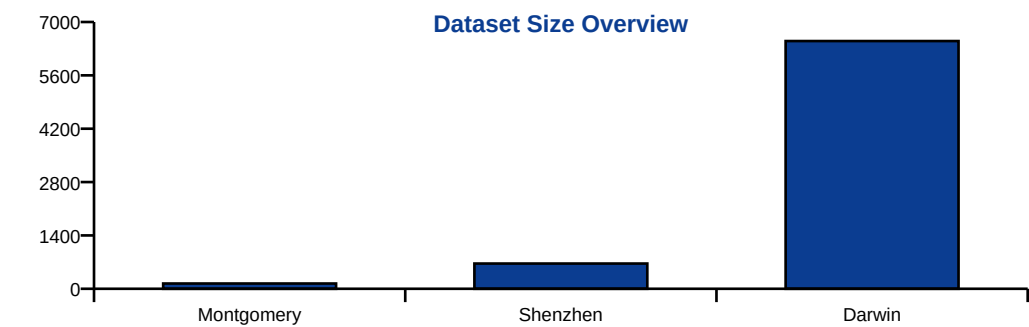
1. Dataset Links

Dataset	Public Link / DOI	Use in Study
Shenzhen Chest X-ray Dataset	https://doi.org/10.6084/m9.figshare.16609978.v1	Frontal CXR images with normal/TB cases and lung masks.
Montgomery County and Darwin Chest X-ray Dataset	https://doi.org/10.17632/8gf9vpkhgy.2	CXR lung segmentation datasets with expert masks, including heterogeneous Darwin cases.

The links above are reproduced from the manuscript data availability statement.

2. Dataset Composition

Dataset	Samples	Clinical / Acquisition Profile	Mask Annotation
Montgomery County	138 CXRs: 80 normal, 58 tuberculosis	Postero-anterior radiographs; 12-bit grayscale PNG; matrix sizes 4020 x 4892 or 4892 x 4020.	Expert-supervised left and right lung binary masks.
Shenzhen	662 CXRs: 326 normal, 336 tuberculosis	Frontal chest X-rays from Shenzhen No. 3 People's Hospital; varying native resolutions around 3K x 3K; adult and pediatric studies.	Lung masks used for segmentation evaluation.
Darwin	6,500 AP/PA frontal CXRs; 517 COVID-19 cases	Highly heterogeneous sources, orientations, and resolutions from 156 x 156 to 5600 x 4700; includes portable/low-quality images.	Pixel-level polygon lung annotations with left and right lung masks.



3. Preprocessing and Splitting

Item	Description
Input preparation	Images and masks are paired by base filename, loaded in 8-bit grayscale using OpenCV, and resized to 128 x 128.
Normalization	CXR intensities are scaled to [0, 1] by dividing by 255. Masks are binarized using threshold 127 and stored as single-channel tensors.
Augmentation	Training-only random horizontal flipping with probability 0.5. Validation and test samples are not augmented.
Data split	General split: 70% training, 15% validation, 15% testing. Montgomery additionally uses 5-fold cross-validation due to limited sample size.
Batching	Mini-batch size of 2 is used in the reported Colab-oriented pipeline to manage GPU memory.

4. Segmentation Performance Statistics

Dataset	Dice	IoU	Precision	Sensitivity	Specificity	AUC	NSD	MSE	PSNR
Montgomery	0.9717	0.9450	0.9751	0.9742	0.9920	0.9934	0.9820	0.0089	22.874
Shenzhen	0.9613	0.9193	0.9696	0.9671	0.9902	0.9925	0.9632	0.0201	21.107
Darwin	0.9540	0.9060	0.9512	0.9390	0.9670	0.9910	0.9290	0.0378	18.528

Montgomery results are reported under 5-fold cross-validation; Shenzhen and Darwin results are reported on their corresponding evaluation splits.

5. Dataset-Specific Observations

Dataset	Observed Behavior
Montgomery	Most stable distribution; high Dice/IoU and low spatial error; errors mainly restricted to thin boundary zones.
Shenzhen	Moderate variability; some boundary errors near rib-adjacent, central, and basal areas due to anatomical and pathological variation.
Darwin	Most challenging domain; wider metric spread, higher false-negative tendency, and more errors near lung bases/peripheral contours due to under-exposure, motion artifacts, and acquisition heterogeneity.

6. Calibration, Thresholds, and Deployment Indicators

Indicator	Montgomery	Shenzhen	Darwin
Approx. optimal threshold	0.467	0.367	0.667
Expected calibration error (ECE)	0.005	0.016	0.031
Inference time	4.40 ms	10.26 ms	8.86 ms
Shared model footprint	41.53 M parameters; 158.41 MB memory	Same	Same

7. Statistical Interpretation

The overall statistical pattern shows strong overlap performance across all datasets, with Montgomery giving the highest Dice and IoU, Shenzhen showing moderate acquisition variability, and Darwin presenting the clearest domain-shift challenge. The increase in MSE and reduction in PSNR from Montgomery to Darwin indicate that segmentation quality is affected by acquisition heterogeneity and artifacts, even though Dice and IoU remain high. The dataset-dependent threshold differences confirm that a single probability threshold may not be optimal under cross-domain deployment.

Source Note

Prepared from the attached manuscript: *Tri_Attention_UNET_28_May_2026_v9.pdf*. Verify URLs and metrics before repository release or journal resubmission.